

Fair Bilevel Optimization with Synthetic Data for Collaborative Classification

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Abstract

We study collaborative classification in the presence of a sensitive attribute and propose a bilevel optimization framework that explicitly targets Equal Opportunity (EO) fairness. Each client generates a synthetic dataset so that classifiers trained on these data remain EO-fair when evaluated on the original distribution. A key observation is that, under joint stratification by the sensitive attribute and the label, minibatch-level fairness constraints aggregate to global EO in expectation. This insight enables the design of a minibatch-friendly algorithm that enforces fairness through a conditional covariance surrogate on qualified individuals and stabilizes training with an Augmented Lagrangian formulation. To improve generalization and avoid overfitting to synthetic data, we incorporate a regularization term. The result is a practical and scalable framework for EO-fair collaborative learning that extends naturally to expressive nonlinear models.

1 Introduction

In collaborative settings, private datasets are distributed across clients, with heterogeneous feature distributions and group compositions. When a sensitive attribute is involved, aggregating such heterogeneous data can result in globally unfair classifiers. To address this and preserve privacy, we seek methods that evaluate fairness on each client’s original data distribution while constructing synthetic subsets that can be safely shared and used for training. Specifically, we study collaborative classification in the presence of a sensitive attribute and propose a bilevel optimization framework that explicitly targets Equal Opportunity (EO) fairness. Each client generates a synthetic dataset so that classifiers trained on these data remain EO-fair when evaluated on the original distribution.

Equal Opportunity requires that, among qualified individuals, the likelihood of receiving a positive prediction does not depend on group membership. Formally, if $S \in \{0, 1\}$ is a binary sensitive attribute, $Y \in \{0, 1\}$ is the true label, and $\hat{Y}$ is the classifier’s prediction, EO is defined as

$$\mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 0) = \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 1). \quad (\text{EO})$$

This condition ensures equality of true positive rates across groups and has become one of the most widely used fairness definitions, since it explicitly conditions on qualification and thereby protects individuals who deserve a favorable decision. In the federated setting, our ultimate goal is a *single global model* that achieves EO across the population represented by all clients. Concretely, fairness means that among qualified individuals ($Y = 1$), the probability of a positive prediction is invariant to S when data from all clients are pooled. Achieving this global guarantee directly is challenging because raw data cannot be shared. Instead, we enforce fairness locally: each client generates synthetic data that is EO-fair with respect to its own original distribution. Through joint

stratification of minibatches by (S, Y) , these local guarantees aggregate (in expectation) to fairness at the global level. In this way, client-level fairness constraints act as safeguards, while server-side aggregation yields a single model that is both accurate and EO-fair.

2 A Fair Bilevel Optimization Approach with Synthetic Data

Suppose we have K clients, each with its original data $\mathcal{D}^o = \{(x_i^o, s_i^o, y_i^o)\}_{i=1}^{N^s}$ (after dropping client's index k). We train a parametric predictor $f_\theta : \mathbb{R}^{n-1} \times \{0, 1\} \rightarrow \mathbb{R}$ by minimizing a regularized empirical risk,

$$\ell_N(\{x_i, s_i, y_i\}, \theta) = \frac{1}{N} \sum_{i=1}^N \ell(y_i, f_\theta(x_i, s_i)) + \lambda_\theta R(\theta), \quad (1)$$

where $\ell(\cdot, \cdot)$ is a convex loss (e.g., logistic/hinge/squared) and $R(\theta) = \frac{1}{2} \|\theta\|_2^2$ is ridge regularization with coefficient $\lambda_\theta > 0$. For instance, with linear $f_\theta(x, s) = [x; s]^\top \theta$ and logistic loss, (19) is ridge-logistic regression.

Let the predicted label be $\hat{Y} = \mathbb{I}\{f_\theta(x, s) \geq \tau\}$ for a fixed threshold τ (e.g., $\tau = 0$ for margin models or τ corresponding to probability 0.5 for calibrated scores). Directly optimizing (EO) is difficult in stochastic settings, so we use the *conditional covariance* surrogate on the $Y=1$ slice:

$$g_{\text{EO}}(\theta) = \frac{1}{|\mathcal{D}_{Y=1}^o|} \sum_{\substack{(x_i^o, s_i^o, y_i^o) \in \mathcal{D}^o \\ y_i^o=1}} (s_i^o - \bar{S}_{Y=1}) f_\theta(x_i^o, s_i^o), \quad \bar{S}_{Y=1} = \frac{1}{|\mathcal{D}_{Y=1}^o|} \sum_{y_i^o=1} s_i^o. \quad (2)$$

For binary S , $g_{\text{EO}}(\theta)$ is proportional to the difference in mean scores between groups on the $Y = 1$ slice, i.e.,

$$g_{\text{EO}}(\theta) = p(1-p) \left(\mathbb{E}[f_\theta(X, S) \mid Y = 1, S = 1] - \mathbb{E}[f_\theta(X, S) \mid Y = 1, S = 0] \right),$$

with $p = \mathbb{P}(S = 1 \mid Y = 1)$. Intuitively, driving $g_{\text{EO}}(\theta)$ to zero aligns the *qualified* score distributions across groups, which controls the TPR gap.

We construct a synthetic dataset $\{(x_i^s, s_i^s, y_i^s)\}_{i=1}^{N^s}$ where (s_i^s, y_i^s) are chosen by stratified subsampling from the client's original (S, Y) distribution. Given a stratified subsampling (s_i^s, y_i^s) , we optimize features $\{x_i^s\}$ through the bilevel program

$$\{x_i^s\}_{i=1}^{N^s} = \arg \min_{\{x_i\}} \ell_{N^o}(\{x_i^o, s_i^o, y_i^o\}, \theta^*) + \lambda_\theta R(\theta^*) \quad (3)$$

$$\text{s.t. } \theta^* \in \arg \min_{\theta} \ell_{N^s}(\{x_i, s_i^s, y_i^s\}, \theta), \quad |g_{\text{EO}}(\theta^*)| \leq \varepsilon_{\text{EO}}, \quad (C)$$

(inner training on synthetic, evaluation & fairness on original)

where $\lambda_\theta > 0$. The outer objective measures performance on original data; the fairness constraint (C) enforces the EO surrogate on original data. Directly solving (21) at full synthetic size is expensive: optimizing over $(n-1) \times N^s$ variables becomes intractable as n or N^s grow. We therefore adopt an iterative minibatch procedure with a *fixed* batch size B per round. At round t , we draw a batch B_t by *jointly stratified* sampling over (S, Y) : each stratum (s, y) contributes $n_{s,y} = B \pi_{s,y}$ points (or randomized rounding) where $\pi_{s,y} = \mathbb{P}(S = s, Y = y)$.

2.1 Augmented Lagrangian EO-Fair Bilevel Optimization

We combine accuracy, regularization, and EO via an Augmented Lagrangian (AL):

$$\mathcal{L}_{\text{outer}}(\theta; \lambda, \rho) = \ell_{\text{task}}(\theta; \mathcal{D}^o) + \lambda_{\theta} R(\theta) + \lambda g_{\text{EO}}(\theta) + \frac{\rho}{2} g_{\text{EO}}(\theta)^2, \quad (4)$$

with multiplier $\lambda \in \mathbb{R}$ and penalty $\rho > 0$. Stochastic updates are

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}_{\text{outer}}(\theta; \lambda, \rho), \quad \lambda \leftarrow \lambda + \rho g_{\text{EO}}(\theta). \quad (5)$$

The linear term encourages feasibility (driving $g_{\text{EO}}(\theta) \rightarrow 0$), while the quadratic term stabilizes the dynamics.

Each client solves the bilevel problem on its local data, producing synthetic data $\mathcal{S}^s$ under the same methodology. Synthetic datasets are shared with a server and aggregated to train a global nonlinear classifier. Because fairness (via AL) and regularization are enforced locally relative to original data, the global model inherits EO properties and improved generalization.

Per-round client bilevel subproblem (EO with Augmented Lagrangian). For client k at round t , the *inner learner* is trained on the tiny synthetic batch $S_{k,t}$:

$$\theta_{k,t} := \arg \min_{\theta} \ell_r(\theta; S_{k,t}). \quad (6)$$

Let $I_{k,t}^{(+)} = \{i \in I_{k,t} : y_i^o = 1\}$ and define the centered sensitive mean on the *positive* slice $\bar{s}_{k,t}^{(+)} = \frac{1}{|I_{k,t}^{(+)}|} \sum_{i \in I_{k,t}^{(+)}} s_i^o$. The EO covariance surrogate on the subsample $I_{k,t}$ is

$$g_{\text{EO}}^{k,t}(\theta) = \frac{1}{|I_{k,t}^{(+)}|} \sum_{i \in I_{k,t}^{(+)}} (s_i^o - \bar{s}_{k,t}^{(+)}) f_{\theta}(x_i^o, s_i^o). \quad (7)$$

We enforce EO via an *augmented Lagrangian* (AL) penalty on $I_{k,t}$:

$$\begin{aligned} \min_{\{x_j^s\}_{j=1}^B} \quad & L_{\text{real}}^{k,t}(\theta_{k,t}; I_{k,t}) + \lambda_{k,t} g_{\text{EO}}^{k,t}(\theta_{k,t}) + \frac{\rho_t}{2} (g_{\text{EO}}^{k,t}(\theta_{k,t}))^2, \\ \text{with} \quad & L_{\text{real}}^{k,t}(\theta; I_{k,t}) = \frac{1}{|I_{k,t}|} \sum_{i \in I_{k,t}} \ell(y_i^o, f_{\theta}(x_i^o, s_i^o)). \end{aligned} \quad (8)$$

(Optionally, an SP term can be added by replacing $I_{k,t}^{(+)}$ with $I_{k,t}$ in (25); in this work we focus on EO only.) The dependence $\theta_{k,t} = \theta_{k,t}(\{x_j^s\})$ is through (24); gradients of (26) w.r.t. $\{x_j^s\}$ are obtained by the chain rule and implicit differentiation of (24). After optimizing (26), the client updates its multiplier as

$$\lambda_{k,t+1} \leftarrow \lambda_{k,t} + \rho_t g_{\text{EO}}^{k,t}(\theta_{k,t}). \quad (9)$$

Algorithm (Client-Centric EO-Fair Synthetic Bilevel Learning)

Algorithm 1 Client-Centric EO via Stratified Minibatches and Augmented Lagrangian

Input: Clients $\{X_o^k\}_{k=1}^K$; rounds T ; batch sizes B (original) and m (synthetic); predictor f_θ ; loss ℓ ; regularizer R ; learning rates η_θ, η_x ; AL params $\{\rho_t\}$; initialize multipliers $\lambda_{k,1}=0$.

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1: Initialize server synthetic set  $S_{1:0} \leftarrow \emptyset$ 
2: for  $t = 1$  to  $T$  do
3:   for each client  $k = 1, \dots, K$  in parallel do
4:     Draw jointly stratified indices  $I_{k,t}$  of size  $B$  from  $X_o^k$  (preserve  $(S, Y)$  proportions)
5:     Form tiny synthetic batch  $S_{k,t} = \{(x_j^s, s_j^s, y_j^s)\}_{j=1}^m$  by fixing  $(s_j^s, y_j^s)$  from  $I_{k,t}$ ; initialize  $x_j^s$ 
6:     repeat ▷ Client bilevel steps (few inner-outer iterations)
7:       Inner step:  $\theta_{k,t} \leftarrow \arg \min_{\theta} \ell_r(\theta; S_{k,t})$  (optimize by SGD/Adam)
8:       EO on subsample: compute  $g_{\text{EO}}^{k,t}(\theta_{k,t})$  via (25)
9:       Outer step on  $x^s$ : minimize (26) w.r.t.  $\{x_j^s\}$  using gradients
10:      AL multiplier: update  $\lambda_{k,t} \leftarrow \lambda_{k,t} + \rho_t g_{\text{EO}}^{k,t}(\theta_{k,t})$  (cf. (27))
11:    until outer improvement below tolerance or max iters
12:    Send  $S_{k,t}$  to server
13:  end for
14:  Server updates  $S_{1:t} \leftarrow S_{1:t-1} \cup (\cup_k S_{k,t})$ 
15:  (Optional) Server trains a global  $f_\theta$  on  $S_{1:t}$  for evaluation/checkpoints
16: end for

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Output: Final cumulative synthetic set $S_{1:T}$ and trained global predictor f_θ

Lemma 1 (Stratified sampling implies (S, Y) -independent batch membership). *Let the dataset have N points with $N_{s,y}$ in stratum $(S = s, Y = y)$ and $\pi_{s,y} = N_{s,y}/N$. Fix a batch size B used at every round. At round t , construct $B_t = \bigcup_{(s,y)} B_t^{(s,y)}$ with $|B_t^{(s,y)}| = n_{s,y}$ drawn uniformly without replacement from each stratum and $n_{s,y} = B \pi_{s,y}$ (or $\mathbb{E}[n_{s,y}] = B \pi_{s,y}$ by randomized rounding). Let U be a uniformly random index and $A_t = \{U \in B_t\}$. Then*

$$\mathbb{P}(A_t \mid S_U = s, Y_U = y) = \mathbb{P}(A_t) \quad \text{for all } (s, y) \in \{0, 1\}^2,$$

i.e., batch membership is independent of (S, Y) (exactly under proportional counts; in expectation under randomized rounding).

Proof. Within stratum (s, y) , $\mathbb{P}(A_t \mid S_U = s, Y_U = y) = n_{s,y}/N_{s,y} = B/N$, while $\mathbb{P}(A_t) = |B_t|/N = B/N$. $\square$

Batch-level EO gap and its surrogate. Define the minibatch EO gap

$$\Delta_t(\theta, \tau) := \mathbb{P}(f_\theta(X, S) \geq \tau \mid Y = 1, S = 0, B_t) - \mathbb{P}(f_\theta(X, S) \geq \tau \mid Y = 1, S = 1, B_t),$$

and the batch-level surrogate

$$g_{\text{EO}}^{(t)}(\theta) = \mathbb{E}[(S - \bar{S}_t^{(+)} f_\theta(X, S) \mid Y = 1, B_t], \quad \bar{S}_t^{(+)} = \mathbb{E}[S \mid Y = 1, B_t]. \quad (10)$$

Lemma 2 (Surrogate controls EO gap on a batch). *Assume that, conditional on $\{Y = 1, B_t, S = s\}$, the score $f_\theta(X, S)$ has a density $q_s^{(t)}$ that is continuous at τ and uniformly bounded by $\lambda_{\text{dens}}^{(t)}$ in a neighborhood of τ . Then*

$$|\Delta_t(\theta, \tau)| \leq \frac{\lambda_{\text{dens}}^{(t)}}{\bar{p}_t(1 - \bar{p}_t)} \left| g_{\text{EO}}^{(t)}(\theta) \right|, \quad \bar{p}_t = \mathbb{P}(S = 1 \mid Y = 1, B_t).$$

Proof sketch. For binary S , $g_{\text{EO}}^{(t)}(\theta) = \bar{p}_t(1 - \bar{p}_t)(\mathbb{E}[f \mid S = 1] - \mathbb{E}[f \mid S = 0])$ on $\{Y=1, B_t\}$. The TPR gap equals $F_0^{(t)}(\tau) - F_1^{(t)}(\tau)$, where $F_s^{(t)}$ is the CDF of f_θ given $\{Y=1, S=s, B_t\}$. By the mean-value theorem and the density bound near τ , the difference in CDFs is bounded by the density constant times the difference in conditional means of f_θ , yielding the stated inequality. $\square$

Theorem 1 (Mini-to-Global EO and surrogate control). *Assume each minibatch B_t is sampled with fixed size B under Lemma 3. Then*

$$\mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 0) - \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 1) = \sum_t \mathbb{P}(B_t) \Delta_t(\theta, \tau).$$

In particular, if $g_{\text{EO}}^{(t)}(\theta) = 0$ for all t , then $\Delta_t(\theta, \tau) = 0$ for all t under Lemma 2's conditions with $\lambda_{\text{dens}}^{(t)} < \infty$, and global EO holds. More generally,

$$\left| \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 0) - \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 1) \right| \leq \sum_t \mathbb{P}(B_t) \frac{\lambda_{\text{dens}}^{(t)}}{\bar{p}_t(1 - \bar{p}_t)} \left| g_{\text{EO}}^{(t)}(\theta) \right|.$$

Proof. By the law of total probability and Lemma 3,

$$\mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = s) = \sum_t \mathbb{P}(B_t) \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = s, B_t).$$

Subtract across $s \in \{0, 1\}$ to get the first identity. The bounds follow by Lemma 2 and the convexity of the absolute value with weights $\mathbb{P}(B_t)$. $\square$

Another less common fairness surrogate is Statistical Parity (SP), which requires

$$\mathbb{P}(\hat{Y} = 1 \mid S = 0) = \mathbb{P}(\hat{Y} = 1 \mid S = 1),$$

ignoring whether individuals are truly qualified. In a mini-batch setting that our ultimate goal is to achieve, SP is often incompatible with EO: if base rates differ, enforcing both typically harms utility. We therefore focus exclusively on EO. Consider two groups, $S = 0$ and $S = 1$, each with ten positives ($Y = 1$). Draw two stratified batches B_1, B_2 of equal size with five positives from each group. If the classifier predicts all positives correctly in each batch, then $\Delta_t(\theta, \tau) = 0$ for $t = 1, 2$. By Theorem 2, the global EO gap is $\sum_t \mathbb{P}(B_t) \Delta_t = 0$, so EO holds globally. Without stratification, one batch might overrepresent one group, and zero batch-level gaps would no longer imply global fairness.

Summary. We replace the costly full-synthetic optimization with a stratified, fixed-size minibatch procedure. The EO surrogate g_{EO} is enforced locally per batch, and Lemmas 3–2 together with Theorem 2 show how these local controls aggregate to global EO. Regularization with coefficient λ_θ prevents overfitting to synthetic data, and the generic formulation accommodates expressive nonlinear predictors f_θ .

3 Addressing Imbalance in EO Estimation

A critical challenge in enforcing Equal Opportunity (EO) arises from *imbalance* in the data. In many applications, the proportion of positives ($Y=1$) differs substantially across groups S , or one group may have very few positive examples overall. Because EO is defined in terms of conditional true positive rates, imbalance can destabilize minibatch estimates of $g_{\text{EO}}(\theta)$ and degrade both fairness and convergence.

3.1 Consequences of Imbalance

When the number of positive examples is highly uneven across groups:

- **Noisy fairness signal.** The covariance surrogate in Eq. (20) relies on $Y=1$ slices. If these are small, minibatch estimates of EO become high-variance.
- **Multiplier instability.** In the augmented Lagrangian formulation, Eq. (23), noisy estimates of $g_{\text{EO}}(\theta)$ can cause oscillations in the multiplier λ .
- **Biased training.** Without correction, majority groups dominate the fairness constraint, leaving minority groups poorly protected.

3.2 Mitigation Strategies

We adopt several techniques to address imbalance while retaining the minibatch-friendly nature of our algorithm:

(i) Joint (S, Y) -stratification with quotas. In addition to stratification over (S, Y) (Eq. (??)), we impose *minimum per-group quotas* of positives in each minibatch:

$$|B_t \cap \{Y = 1, S = s\}| \geq k_{\min}, \quad s \in \{0, 1\}.$$

If quotas cannot be met directly, we maintain a rolling buffer of positive examples per group and sample from it to fill batches. This ensures that each EO estimate is computed on a non-degenerate slice.

(ii) Importance reweighting. Let $\pi_s = \mathbb{P}(S = s \mid Y = 1)$ denote group proportions among positives. We assign inverse-propensity weights

$$w_i = \frac{1}{\pi_{s_i}}, \quad \text{for all } y_i = 1,$$

with clipping to $w_{\max}$ to control variance. The weighted EO estimator is

$$\hat{g}_{\text{EO}}^{(t)}(\theta) = \frac{\sum_{i \in B_t, y_i=1} w_i (s_i - \bar{S}_w) f_{\theta}(x_i)}{\sum_{i \in B_t, y_i=1} w_i},$$

where $\bar{S}_w$ is the weighted group mean on $Y=1$. This corrects bias from unbalanced group sizes.

(iii) Variance reduction. To further reduce stochastic noise, we maintain an exponential moving average (EMA) of EO estimates:

$$\tilde{g}_t = (1 - \beta)\tilde{g}_{t-1} + \beta\hat{g}_{\text{EO}}^{(t)}(\theta_t), \quad \beta \in [0.05, 0.25],$$

and use $\tilde{g}_t$ in the augmented Lagrangian instead of raw $\hat{g}_{\text{EO}}^{(t)}$. Gradient clipping is also applied to avoid rare spikes when minority counts are very small.

(iv) Optional augmentation. In extreme imbalance settings, we generate synthetic augmentations of positive examples within each group (e.g., mixup in feature space). These augmented points are used only to stabilize EO estimation, not to alter the fairness target.

3.3 Resulting Benefits

Together, these techniques ensure that:

- EO estimates remain unbiased for the global conditional covariance.
- Variance is controlled, preventing instability in multiplier updates.
- Minority groups receive consistent protection from fairness violations.

This imbalance-robust extension preserves the local-to-global EO guarantee (Theorem 2) while improving stability and fairness in heterogeneous collaborative environments.

4 EO Enforcement via TPR Gap

4.1 Equal Opportunity Through True Positive Rates

By definition, Equal Opportunity requires that true positive rates (TPRs) are equal across groups. For group $s \in \{0, 1\}$, we define

$$\text{TPR}_s(\theta, \tau) = \mathbb{P}(f_\theta(X) \geq \tau \mid Y = 1, S = s), \quad (11)$$

where $f_\theta(X)$ is the classifier score and τ is the decision threshold. EO fairness is satisfied if

$$\text{TPR}_0(\theta, \tau) = \text{TPR}_1(\theta, \tau).$$

4.2 Empirical Estimation

On a finite dataset, the group-specific TPRs are estimated as

$$\widehat{\text{TPR}}_s(\theta, \tau) = \frac{1}{|\mathcal{I}_s|} \sum_{i \in \mathcal{I}_s} \mathbf{1}\{f_\theta(x_i) \geq \tau\}, \quad \mathcal{I}_s = \{i : y_i = 1, s_i = s\}. \quad (12)$$

This estimator measures the fraction of truly positive individuals in group s that the classifier predicts as positive.

4.3 Differentiable Surrogate via Sigmoid Smoothing

The indicator in (12) is non-differentiable, which prevents gradient-based optimization. To overcome this, we approximate the step function with a smooth sigmoid:

$$\mathbf{1}\{f_\theta(x_i) \geq \tau\} \approx \sigma(\alpha \cdot (f_\theta(x_i) - \tau)), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (13)$$

where $\alpha > 0$ controls the sharpness of the approximation. Larger values of α make the sigmoid closer to the hard indicator, while smaller values yield smoother gradients.

The resulting differentiable surrogate is

$$\widetilde{\text{TPR}}_s(\theta, \tau) = \frac{1}{|\mathcal{I}_s|} \sum_{i \in \mathcal{I}_s} \sigma(\alpha \cdot (f_\theta(x_i) - \tau)). \quad (14)$$

4.4 Fairness Penalty Based on TPR Gap

Using the smooth TPR estimates, we define an EO fairness penalty as the squared TPR gap:

$$\Phi_{\text{EO}}(\theta, \tau) = \left(\widetilde{\text{TPR}}_0(\theta, \tau) - \widetilde{\text{TPR}}_1(\theta, \tau) \right)^2. \quad (15)$$

This penalty is differentiable in both θ and τ . Optimizing over (θ, τ) balances predictive performance with fairness.

4.5 Joint Optimization of Model and Threshold

Unlike covariance-based surrogates, the TPR-gap formulation explicitly incorporates the decision threshold. This opens two possibilities:

- **Fixed threshold:** τ is set to a default value (e.g., 0 for logits).
- **Learned threshold:** τ is treated as an additional optimization variable, updated jointly with θ .

Jointly learning τ aligns training with the deployed operating point, ensuring that the final model is EO-fair at the threshold used in practice.

4.6 Objective Function

The regularized training objective becomes

$$\min_{\theta, \tau} \ell_{\text{task}}(\theta) + R(\theta) + \rho \Phi_{\text{EO}}(\theta, \tau), \quad (16)$$

where ℓ_{task} is the prediction loss, $R(\theta)$ the regularization term, and $\rho > 0$ a fairness weight.

4.7 Discussion

The TPR-gap formulation captures EO directly and is interpretable, since the fairness penalty is exactly the reported metric. However, it introduces additional complexity: minibatch estimates of $\widetilde{\text{TPR}}_s$ can be noisy under imbalance, and optimization requires tuning α to trade off smoothness and accuracy of the surrogate. In contrast, the covariance surrogate is simpler and more stable but only indirectly enforces EO. In practice, one may use the covariance-based method for scalable training and apply TPR-based penalties or threshold adjustment as a refinement step.

References

- [1] Moritz Hardt, Eric Price, and Nati Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.

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Equal Opportunity requires that, among qualified individuals, the likelihood of receiving a positive prediction does not depend on group membership. Formally, if $S \in \{0, 1\}$ is a binary sensitive attribute, $Y \in \{0, 1\}$ is the true label, and $\hat{Y}$ is the classifier’s prediction, EO is defined as

$$\mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 0) = \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 1). \quad (17)$$

This condition ensures equality of true positive rates across groups and has become one of the most widely used fairness definitions, since it explicitly conditions on qualification and thereby protects individuals who deserve a favorable decision. By contrast, Statistical Parity (SP) requires

$$\mathbb{P}(\hat{Y} = 1 \mid S = 0) = \mathbb{P}(\hat{Y} = 1 \mid S = 1), \quad (18)$$

ignoring whether individuals are truly qualified. While easy to state, SP is often incompatible with EO: if the base rates of qualification differ between groups, enforcing both simultaneously is generally impossible without harming predictive performance. For this reason, we focus exclusively on EO in our framework.

In the federated setting, our ultimate goal is to obtain a *single global model* that achieves EO fairness across the entire population represented by all clients. Concretely, fairness here means that among qualified individuals ($Y = 1$), the probability of receiving a positive prediction should not depend on the sensitive attribute S when data from all clients are pooled together. Achieving this global guarantee directly is challenging because raw data cannot be shared across clients. Instead, we enforce fairness locally: each client generates synthetic data that is EO-fair with respect to its own original distribution. Through joint stratification of minibatches by (S, Y) , these local guarantees aggregate in expectation to fairness at the global level. In this way, client-level fairness constraints act as safeguards, while the server’s aggregation process yields a single model that is both accurate and EO-fair across the federated population.

Suppose we have K clients, each with an original dataset $\mathcal{D}^o = \{(x_i^o, s_i^o, y_i^o)\}$ consisting of features $x_i^o \in \mathbb{R}^{n-1}$, a sensitive attribute $s_i^o \in \{0, 1\}$, and a label $y_i^o \in \{0, 1\}$. We aim to develop a global predictor f_θ that is accurate while adhering to EO in a bilevel optimization framework. In particular, we require the inner problem to train a parametric predictor $f_\theta : \mathbb{R}^{n-1} \times \{0, 1\} \rightarrow \mathbb{R}$ by minimizing a regularized empirical risk,

$$\ell_{N,r}(\{x_i, s_i, y_i\}_{i=1}^N, \theta) = \frac{1}{N} \sum_{i=1}^N \ell(y_i, f_\theta(x_i, s_i)) + \lambda_\theta R(\theta), \quad (19)$$

where $\ell(\cdot, \cdot)$ is a convex loss (e.g., logistic, hinge, or squared loss) and $R(\theta)$ is a regularizer. For instance, with linear $f_\theta(x, s) = [x; s]^\top \theta$ and logistic loss, this reduces to the standard ridge-regularized logistic regression objective.

Since directly imposing (17) is difficult, we use the *conditional covariance* surrogate on the $Y=1$ slice:

$$g_{\text{EO}}(\theta) = \frac{1}{|\mathcal{D}_{Y=1}^o|} \sum_{(x_i^o, s_i^o, y_i^o) \in \mathcal{D}^o, y_i^o=1} (s_i^o - \bar{S}_{Y=1}) f_\theta(x_i^o), \quad (20)$$

where $\bar{S}_{Y=1}$ is the mean of S over $\{i : y_i^o = 1\}$. Thus, our goal is to construct a synthetic dataset $\{(x_i^s, s_i^s, y_i^s)\}_{i=1}^{N^s}$ where (s_i^s, y_i^s) are chosen by stratified subsampling from the original distribution. After stratification with respect to (S, Y) , the synthetic features are obtained by solving the bilevel program

$$\{x_i^s\}_{i=1}^{N^s} = \arg \min_{\{x_i\}_{i=1}^{N^s}} \ell_{N^o, r}(\{x_i^o, s_i^o, y_i^o\}_{i=1}^{N^o}, \theta_{N^s}(\{x_i, s_i^s, y_i^s\}_{i=1}^{N^s})) + \lambda_\theta R(\theta) \quad (21)$$

$$\text{s.t. } |\text{Cov}(s^o, f_{\theta_{N^s}}(x^o, s^o) \mid y^o = 1)| \leq \varepsilon_{\text{EO}}, \quad (3c)$$

where the fairness constraint enforces that, among qualified individuals ($y^o = 1$), the model's predictions are approximately uncorrelated with the sensitive attribute, serving as a tractable surrogate for EO.

However, directly solving (21) is computationally expensive: each synthetic dataset spans an ambient space of dimension $(n - 1)$ across N^s synthetic points, resulting in an optimization problem of size $(n - 1) \times N^s$. This quickly becomes intractable as either the feature dimension or the number of synthetic samples grows. To overcome this, we will introduce an iterative minibatch-based procedure with a fixed batch size B at each round. A key technical component is *joint stratification* with respect to both the sensitive attribute and the label, which guarantees that batch membership is independent of (S, Y) .

Lemma 3 (Stratified sampling implies (S, Y) -independent batch membership). *Let the dataset have N points and let $N_{s,y}$ denote the number of points in stratum $(S = s, Y = y)$, with proportions $\pi_{s,y} = N_{s,y}/N$. Fix a batch size B that is used at every round. At round t , construct B_t by drawing exactly $n_{s,y}$ points without replacement from each stratum (s, y) , and take $B_t = \bigcup_{(s,y)} B_t^{(s,y)}$ with $|B_t^{(s,y)}| = n_{s,y}$. Assume proportional allocation $n_{s,y} = B \pi_{s,y}$ whenever $B \pi_{s,y} \in \mathbb{N}$, or else use randomized rounding so that $\mathbb{E}[n_{s,y}] = B \pi_{s,y}$.*

Let U be a drawback, uniformly random index from $\{1, \dots, N\}$; write $A_t = \{U \in B_t\}$ for the event that U is included in batch B_t . Then

$$\mathbb{P}(A_t \mid S_U = s, Y_U = y) = \mathbb{P}(A_t) \quad \text{for all } (s, y) \in \{0, 1\}^2,$$

i.e., batch membership is independent of the pair (S, Y) . The equality holds exactly under proportional allocation, and in expectation under randomized rounding.

Proof. Condition on stratum $(S_U = s, Y_U = y)$. Because selection within stratum (s, y) is uniform without replacement,

$$\mathbb{P}(A_t \mid S_U = s, Y_U = y) = \frac{n_{s,y}}{N_{s,y}}.$$

If proportional allocation is exact, $n_{s,y} = B \pi_{s,y} = B N_{s,y}/N$, hence

$$\mathbb{P}(A_t \mid S_U = s, Y_U = y) = \frac{B}{N}.$$

On the other hand, since $|B_t| = B$,

$$\mathbb{P}(A_t) = \frac{|B_t|}{N} = \frac{B}{N}.$$

Thus $\mathbb{P}(A_t \mid S_U = s, Y_U = y) = \mathbb{P}(A_t)$. With randomized rounding, equality holds in expectation. $\square$

This immediately implies the following result.

Theorem 2 (Mini-to-Global EO). *Assume each minibatch B_t is sampled with fixed size B under the stratification scheme of Lemma 3. Let*

$$\Delta_t = \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 0, B_t) - \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 1, B_t)$$

denote the EO gap on minibatch B_t . Then the global EO gap satisfies

$$\mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 0) - \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 1) = \sum_t \mathbb{P}(B_t) \Delta_t.$$

In particular, if $\Delta_t = 0$ for all t , then the global EO constraint holds exactly.

Proof. By the law of total probability, for any $s \in \{0, 1\}$,

$$\mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = s) = \sum_t \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = s, B_t) \mathbb{P}(B_t \mid Y = 1, S = s).$$

By Lemma 3, $\mathbb{P}(B_t \mid Y = 1, S = s) = \mathbb{P}(B_t)$. Substituting gives

$$\mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = s) = \sum_t \mathbb{P}(B_t) \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = s, B_t).$$

Subtracting the two group expressions yields

$$\mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 0) - \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 1) = \sum_t \mathbb{P}(B_t) \Delta_t,$$

which proves the claim. $\square$

To see why we are not imposing (17) and (18) or their surrogates simultaneously, consider two groups, $S = 0$ and $S = 1$, each with ten positive ($Y = 1$) examples. Suppose we draw two stratified minibatches B_1 and B_2 , each containing five positives from each group. If a classifier predicts correctly on all positives in each minibatch, then within each B_t we have $\mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 0, B_t) = \mathbb{P}(\hat{Y} = 1 \mid Y = 1, S = 1, B_t) = 1$. By stratification, the global EO gap equals the weighted sum of minibatch gaps, which in this case is zero. Hence global EO is satisfied. Without stratification, one minibatch might contain mostly positives from $S = 0$ and few from $S = 1$, in which case equal treatment within the batch would not guarantee fairness globally. This example highlights why stratification is essential for lifting minibatch-level EO constraints to the full collaborative model.

In summary, our approach replaces the expensive optimization over entire synthetic datasets with an iterative, minibatch-based procedure. Stratification ensures that enforcing EO locally propagates to global fairness, while the bilevel structure allows each client to produce synthetic data that balances accuracy and fairness when aggregated.

5 Augmented Lagrangian EO-Fair Bilevel Optimization

We combine accuracy, regularization, and EO via an Augmented Lagrangian (AL):

$$\mathcal{L}_{\text{outer}}(\theta; \lambda, \rho) = \ell_{\text{task}}(\theta; \mathcal{D}^o) + \lambda_{\theta} R(\theta) + \lambda g_{\text{EO}}(\theta) + \frac{\rho}{2} g_{\text{EO}}(\theta)^2, \quad (22)$$

with multiplier $\lambda \in \mathbb{R}$ and penalty $\rho > 0$. Stochastic updates are

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}_{\text{outer}}(\theta; \lambda, \rho), \quad \lambda \leftarrow \lambda + \rho g_{\text{EO}}(\theta). \quad (23)$$

The linear term encourages feasibility (driving $g_{\text{EO}}(\theta) \rightarrow 0$), while the quadratic term stabilizes the dynamics.

Each client solves the bilevel problem on its local data, producing synthetic data $\mathcal{S}^s$ under the same methodology. Synthetic datasets are shared with a server and aggregated to train a global nonlinear classifier. Because fairness (via AL) and regularization are enforced locally relative to original data, the global model inherits EO properties and improved generalization.

Per-round client bilevel subproblem (EO with Augmented Lagrangian). For client k at round t , the *inner learner* is trained on the tiny synthetic batch $S_{k,t}$:

$$\theta_{k,t} := \arg \min_{\theta} \ell_r(\theta; S_{k,t}). \quad (24)$$

Let $I_{k,t}^{(+)} = \{i \in I_{k,t} : y_i^o = 1\}$ and define the centered sensitive mean on the *positive* slice $\bar{s}_{k,t}^{(+)} = \frac{1}{|I_{k,t}^{(+)}|} \sum_{i \in I_{k,t}^{(+)}} s_i^o$. The EO covariance surrogate on the subsample $I_{k,t}$ is

$$g_{\text{EO}}^{k,t}(\theta) = \frac{1}{|I_{k,t}^{(+)}|} \sum_{i \in I_{k,t}^{(+)}} (s_i^o - \bar{s}_{k,t}^{(+)}) f_{\theta}(x_i^o, s_i^o). \quad (25)$$

We enforce EO via an *augmented Lagrangian* (AL) penalty on $I_{k,t}$:

$$\begin{aligned} \min_{\{x_j^s\}_{j=1}^m} \quad & L_{\text{real}}^{k,t}(\theta_{k,t}; I_{k,t}) + \lambda_{k,t} g_{\text{EO}}^{k,t}(\theta_{k,t}) + \frac{\rho_t}{2} (g_{\text{EO}}^{k,t}(\theta_{k,t}))^2, \\ \text{with} \quad & L_{\text{real}}^{k,t}(\theta; I_{k,t}) = \frac{1}{|I_{k,t}|} \sum_{i \in I_{k,t}} \ell(y_i^o, f_{\theta}(x_i^o, s_i^o)). \end{aligned} \quad (26)$$

(Optionally, an SP term can be added by replacing $I_{k,t}^{(+)}$ with $I_{k,t}$ in (25); in this work we focus on EO only.) The dependence $\theta_{k,t} = \theta_{k,t}(\{x_j^s\})$ is through (24); gradients of (26) w.r.t. $\{x_j^s\}$ are obtained by the chain rule and implicit differentiation of (24). After optimizing (26), the client updates its multiplier as

$$\lambda_{k,t+1} \leftarrow \lambda_{k,t} + \rho_t g_{\text{EO}}^{k,t}(\theta_{k,t}). \quad (27)$$

Algorithm (Client-Centric EO-Fair Synthetic Bilevel Learning)

Algorithm 2 Client-Centric EO via Stratified Minibatches and Augmented Lagrangian

Input: Clients $\{X_o^k\}_{k=1}^K$; rounds T ; batch sizes B (original) and m (synthetic); predictor f_{θ} ; loss ℓ ; regularizer R ; learning rates η_{θ}, η_x ; AL params $\{\rho_t\}$; initialize multipliers $\lambda_{k,1} = 0$.

- 1: Initialize server synthetic set $S_{1:0} \leftarrow \emptyset$
- 2: **for** $t = 1$ **to** T **do**
- 3: **for each** client $k = 1, \dots, K$ **in parallel do**
- 4: Draw jointly stratified indices $I_{k,t}$ of size B from X_o^k (preserve (S, Y) proportions)
- 5: Form tiny synthetic batch $S_{k,t} = \{(x_j^s, s_j^s, y_j^s)\}_{j=1}^m$ by fixing (s_j^s, y_j^s) from $I_{k,t}$; initialize x_j^s
- 6: **repeat** ▷ Client bilevel steps (few inner-outer iterations)
- 7: **Inner step:** $\theta_{k,t} \leftarrow \arg \min_{\theta} \ell_r(\theta; S_{k,t})$ (optimize by SGD/Adam)
- 8: **EO on subsample:** compute $g_{\text{EO}}^{k,t}(\theta_{k,t})$ via (25)
- 9: **Outer step on x^s :** minimize (26) w.r.t. $\{x_j^s\}$ using gradients
- 10: **AL multiplier:** update $\lambda_{k,t} \leftarrow \lambda_{k,t} + \rho_t g_{\text{EO}}^{k,t}(\theta_{k,t})$ (cf. (27))
- 11: **until** outer improvement below tolerance or max iters
- 12: Send $S_{k,t}$ to server
- 13: **end for**
- 14: Server updates $S_{1:t} \leftarrow S_{1:t-1} \cup (\cup_k S_{k,t})$
- 15: (Optional) Server trains a global f_{θ} on $S_{1:t}$ for evaluation/checkpoints
- 16: **end for**

Output: Final cumulative synthetic set $S_{1:T}$ and trained global predictor f_{θ}

Remarks. (i) Joint (S, Y) -stratification ensures minibatch EO aggregates to global EO in expectation (Theorem 2). (ii) The formulation is *model-agnostic*: f_θ can be linear or nonlinear (e.g., neural nets, kernels). (iii) The AL update (27) stabilizes constraint satisfaction more effectively than a penalty-only term. (iv) Complexity is controlled via $m \ll B \ll N_k^o$, replacing an $(n - 1) \times N^s$ -sized synthetic optimization with inexpensive client-side micro-batches.

When group-based rates differ, enforcing both SP and EO simultaneously can require extreme false-positive rates and damage accuracy; SP also ignores qualification. Focusing on EO aligns fairness with qualified individuals and avoids structural conflicts. Joint (S, Y) -stratification ensures minibatch estimates reflect the global joint distribution; Theorem 2 then yields a clean local-to-global guarantee for EO in expectation. The Augmented Lagrangian stabilizes fairness constraint satisfaction during stochastic training, while regularization improves generalization and prevents overfitting to synthetic data.

5.1 Universum-Enhanced Synthetic Bilevel Learning for Imbalance

Motivation. In imbalanced settings, the minority class is underrepresented, and ERM tends to skew the decision boundary toward the minority, harming recall. *Universum* points—samples that do not belong to either class of interest but constrain confidence in regions of uncertainty—act as a structural regularizer that flattens overly confident scores near the boundary and reduces bias induced by imbalance.

Universum set and neutrality. At round t and client k , let $\mathcal{U}_{k,t} = \{u_j\}_{j=1}^{m_U}$ denote a batch of Universum points. These points are *unlabeled* (no Y) and are intended to be neutral with respect to S . We adopt one of the following neutrality conditions:

$$(i) \ S \perp U \text{ and } \mathbb{P}(S = 1 \mid U) = \mathbb{P}(S = 1) \quad \text{or} \quad (ii) \ \#\{u \in \mathcal{U}_{k,t} : S(u) = 1\} = \#\{u \in \mathcal{U}_{k,t} : S(u) = 0\}. \quad (28)$$

In practice, (ii) is enforced by drawing Universum points with *joint stratification over S* (and optionally over covariate bins) or by explicit balancing.

Universum penalty. We penalize overly confident scores on Universum points via a symmetric *margin-squeezing* penalty $\phi : \mathbb{R} \rightarrow \mathbb{R}_+$ applied to $f_\theta(u)$:

$$\phi_\varepsilon^{\text{hinge}}(z) = \max\{|z| - \varepsilon, 0\}, \quad \phi_\varepsilon^{\text{sq}}(z) = (\max\{|z| - \varepsilon, 0\})^2, \quad (29)$$

with a small tolerance $\varepsilon \geq 0$ defining the *Universum margin band*. The Universum loss on $\mathcal{U}_{k,t}$ is

$$L_U^{k,t}(\theta; \mathcal{U}_{k,t}) := \frac{1}{m_U} \sum_{u \in \mathcal{U}_{k,t}} \phi(f_\theta(u)). \quad (30)$$

Inner learner with Universum. Augment the inner problem (24) to incorporate Universum points with weight $\lambda_U > 0$:

$$\theta_{k,t} := \arg \min_{\theta} \left\{ \ell_m(\theta; S_{k,t}) + \lambda_\theta R(\theta) + \lambda_U L_U^{k,t}(\theta; \mathcal{U}_{k,t}) \right\}. \quad (31)$$

Here λ_U controls the strength of margin squeezing near the boundary and is the *only new weight* introduced for Universum; we keep the AL parameters (λ, ρ) unchanged.

Outer objective (unchanged structure, richer inner dependence). The outer step remains the same as in (26), but $\theta_{k,t}$ now depends on $\mathcal{U}_{k,t}$ through (31). Gradients w.r.t. synthetic features $\{x_j^s\}$ still follow by the chain rule and implicit differentiation.

EO neutrality under Universum. The Universum term does not directly affect the EO surrogate, which is defined on the $Y = 1$ slice. Under the neutrality condition (28), the Universum modification leaves the *expected* EO surrogate unchanged while reducing the variance of scores around the boundary.

Lemma 4 (Universum neutrality for EO surrogate). *Assume (28) and that f_θ is trained by (31) with ϕ symmetric: $\phi(z) = \phi(-z)$. Then, on the qualified slice $\{Y = 1\}$,*

$$\mathbb{E}[g_{\text{EO}}^{(t)}(\theta_{k,t})] = \mathbb{E}[g_{\text{EO}}^{(t)}(\theta_{k,t}^\circ)],$$

where $\theta_{k,t}^\circ$ is the optimizer of the inner problem with $\lambda_U = 0$. Moreover, the conditional variance of scores near the threshold τ is weakly reduced by the Universum penalty, tightening the bound in Lemma 2 via a smaller effective density constant $\lambda_{\text{dens}}^{(t)}$ in a neighborhood of τ .

Proof sketch. Symmetry of ϕ implies equal pressure toward the margin from both sides of the decision surface. Under (28), this pressure is balanced across S , so the first-order bias in the difference of conditional means on the $Y=1$ slice cancels in expectation. The shrinkage of $|f_\theta|$ around τ reduces local score dispersion, which lowers local density peaks and tightens the surrogate bound. $\square$

Imbalance-aware Universum sampling. Let $\alpha \in [0, 1]$ trade off two sources of Universum points per client-round:

$$\mathcal{U}_{k,t} = \mathcal{U}_{k,t}^{\text{bdry}} \cup \mathcal{U}_{k,t}^{\text{lowdens}},$$

where (i) $\mathcal{U}^{\text{bdry}}$ are small perturbations of real samples projected toward the current decision boundary (virtual adversarial or gradient-normal perturbations), and (ii) $\mathcal{U}^{\text{lowdens}}$ are drawn from estimated low-density regions of $p(x | S)$ (e.g., via kernel density or distance-to- k NN heuristics). We enforce (ii) to be S -balanced per (28) and keep (Y) undefined.

Choice of ϕ . We recommend $\phi_\varepsilon^{\text{sq}}$ in (29) for smooth gradients; set ε at the operating threshold margin (e.g., $\varepsilon = 0$ for logits).

Hyperparameters. All hyperparameters are named with λ :

$$\lambda_\theta \text{ (ridge), } \lambda \text{ (AL multiplier), } \rho \text{ (AL penalty), } \lambda_U \text{ (Universum weight).}$$

6 Methodology

We propose a client-centric bilevel optimization framework that combines (i) *synthetic data generation*, (ii) *Equal Opportunity (EO) fairness* via conditional covariance surrogates, and (iii) *imbalance-aware Universum regularization*. Each client locally enforces fairness and robustness relative to its original distribution, while sharing only synthetic points with the server. A global classifier trained on these synthetic sets inherits both EO-fairness and improved minority-class protection.

6.1 Preliminaries and Notation

suppose each client $k \in \{1, \dots, K\}$ owns a private dataset

$$\mathcal{D}_k^o = \{(x_i^o, s_i^o, y_i^o)\}_{i=1}^{N_k^o}, \quad x_i^o \in \mathbb{R}^d, \quad s_i^o \in \{0, 1\}, \quad y_i^o \in \{0, 1\},$$

where s is the sensitive attribute and y the label, where y distribution is potentially imbalanced, that is, we have majority and minority classes. We train a predictor $f_\theta : \mathbb{R}^d \times \{0, 1\} \rightarrow \mathbb{R}$ with parameters θ , predicting

$$\hat{Y} = I\{f_\theta(x, s) \geq \tau\}.$$

For a given dataset $\mathcal{D} = \{(x_i, s_i, y_i)\}_{i=1}^N$, the generic empirical risk (task loss plus ridge penalty) is

$$\ell_{\text{task,reg}}(\theta; \mathcal{D}) = \frac{1}{N} \sum_{(x,s,y) \in \mathcal{D}} \ell(y, f_\theta(x, s)) + \frac{\lambda_\theta}{2} \|\theta\|_2^2, \quad (32)$$

with convex loss ℓ (e.g., logistic).

6.2 EO Fairness Surrogate

Equal Opportunity (EO) requires equality of true positive rates across groups S . Following (20), we use the conditional covariance surrogate on the $Y = 1$ slice:

$$g_{\text{EO}}(\theta; \mathcal{D}) = \frac{1}{|\mathcal{D}_{Y=1}|} \sum_{(x,s,1) \in \mathcal{D}} (s - \bar{S}_{Y=1}) f_\theta(x, s), \quad \bar{S}_{Y=1} = \frac{1}{|\mathcal{D}_{Y=1}|} \sum_{(x,s,1) \in \mathcal{D}} s. \quad (33)$$

Driving $g_{\text{EO}}(\theta)$ toward zero aligns the mean scores of qualified individuals across groups.

6.3 Synthetic Data Bilevel Formulation

Each client constructs a synthetic dataset

$$\mathcal{D}_k^s = \{(x_i^s, s_i^s, y_i^s)\}_{i=1}^{N_k^s},$$

where (s_i^s, y_i^s) are chosen by joint stratification from the (S, Y) distribution of $\mathcal{D}_k^o$, and features x_i^s are optimized via the bilevel program

$$\begin{aligned} \{x_i^s\} &= \arg \min_{\{x_i\}} \left\{ \ell_{N_k^o}(\theta^*; \mathcal{D}_k^o) \right\} \\ \text{s.t. } \theta^* &\in \arg \min_{\theta} \left\{ \ell_{N_k^s}(\theta; \{(x_i, s_i^s, y_i^s)\}) \right\}, \quad |g_{\text{EO}}(\theta^*; \mathcal{D}_k^o)| \leq \varepsilon_{\text{EO}}. \end{aligned} \quad (34)$$

The outer objective evaluates performance on real data; the inner problem trains on synthetic data with fairness checked on real data.

6.4 Minibatch Stratification and Augmented Lagrangian

To scale, we replace (34) by iterative minibatch optimization. Each round t draws a minibatch $B_{k,t}$ of size B stratified over (S, Y) ; synthetic batch $S_{k,t}$ of size m is generated accordingly.

EO is enforced via an Augmented Lagrangian:

$$\mathcal{L}_{\text{AL}}(\theta; \lambda, \rho) = \ell_{\text{task,reg}}(\theta; B_{k,t}) + \lambda g_{\text{EO}}(\theta; B_{k,t}) + \frac{\rho}{2} g_{\text{EO}}(\theta; B_{k,t})^2, \quad (35)$$

with updates

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}_{\text{AL}}, \quad \lambda \leftarrow \lambda + \rho g_{\text{EO}}(\theta; B_{k,t}).$$

6.5 Universum Regularization for Imbalance

In highly imbalanced scenarios, the decision boundary skews toward the minority class, reducing its recall. To counteract this, we introduce *Universum points* as synthetic, non-class samples that act as a *shield* for the minority class.

At round t , each client constructs a Universum set $\mathcal{U}_{k,t} = \{u_j\}_{j=1}^{m_U}$. These points have no labels Y but are assigned to the *minority side* of the hyperplane. Privacy is preserved since Universum points are either external (public auxiliary data) or synthetic perturbations of local data, never raw samples.

Asymmetric Universum penalty. To enforce shielding, we penalize Universum points only if they fall into the *majority side*:

$$L_U^{k,t}(\theta; \mathcal{U}_{k,t}) = \frac{1}{m_U} \sum_{u \in \mathcal{U}_{k,t}} \max\{f_\theta(u) - \tau, 0\}. \quad (36)$$

Thus, Universum points are encouraged to stay on the minority side of the margin.

Inner learner with Universum. The Universum-augmented inner problem is

$$\theta_{k,t} = \arg \min_{\theta} \left\{ \ell_m(\theta; S_{k,t}) + \lambda_\theta R(\theta) + \lambda_U L_U^{k,t}(\theta; \mathcal{U}_{k,t}) \right\}, \quad (37)$$

where $\lambda_U > 0$ controls the strength of Universum shielding.

Effect on the hyperplane. The asymmetric loss (36) shifts feasible hyperplanes away from the minority region, counteracting the pull of the majority class. In expectation, Universum shielding preserves EO fairness (since Y is undefined) while improving minority recall.

6.6 Algorithm Overview

Each client executes the following steps per round:

1. Draw stratified minibatch $B_{k,t}$ from $\mathcal{D}_k^o$.
2. Initialize synthetic batch $S_{k,t}$ with fixed (s, y) strata.
3. Generate Universum set $\mathcal{U}_{k,t}$ (balanced across S , forced toward minority side).
4. Solve inner problem (37) to update $\theta_{k,t}$.
5. Evaluate EO surrogate $g_{\text{EO}}^{k,t}(\theta_{k,t})$ and update multiplier $\lambda_{k,t}$.
6. Optimize $\{x_j^s\}$ of $S_{k,t}$ by outer loop gradients.
7. Send $S_{k,t}$ to server.

The server aggregates synthetic batches $\{S_{k,t}\}$ across clients and rounds, and trains a global classifier f_θ .

6.7 Hyperparameters

The framework uses four hyperparameters, all denoted by λ with subscripts:

λ_θ (ridge regularization), λ (AL multiplier), ρ (AL penalty), λ_U (Universum penalty).

Typical defaults: $\lambda_\theta \in [10^{-4}, 10^{-2}]$, $\rho \in [0.1, 1]$, and $\lambda_U \in [0.01, 0.2]$ depending on imbalance severity.

6.8 Y-Universum Positives for Imbalance (Balanced in S)

Rationale. Let M and m denote the number of majority and minority class samples (w.r.t. Y) in a client's original dataset $\mathcal{D}_k^o$. To correct class imbalance without distorting EO fairness across the sensitive attribute S , we generate $\Delta := M - m$ *pseudo-positive* points (we call them *Y-Universum positives*) with label $y = 1$ and assign their sensitive attribute *balanced across groups*:

$$\underbrace{\#\{(x_u^s, s_u^s, y_u^s) \in \mathcal{U}_{k,t}^{(Y)} : y_u^s = 1, s_u^s = 0\}}_{= \Delta/2} = \underbrace{\#\{(x_u^s, s_u^s, y_u^s) \in \mathcal{U}_{k,t}^{(Y)} : y_u^s = 1, s_u^s = 1\}}_{= \Delta/2}.$$

These points repair the Y -imbalance (class prior skew) while leaving the EO surrogate unbiased in expectation because the $Y=1$ additions are split evenly across S .

Construction. At round t , client k forms a set $\mathcal{U}_{k,t}^{(Y)} = \{(x_u^s, s_u^s, 1)\}_{u=1}^{\Delta}$ with:

1. **Labels:** $y_u^s \equiv 1$ (qualified/pseudo-positive).
2. **Sensitive attribute:** exactly half with $s_u^s = 0$ and half with $s_u^s = 1$.
3. **Features:** (x_u^s) generated locally by one of:
 - *Boundary interpolation:* convex combinations of minority positives and nearby points projected toward the current decision boundary.
 - *Low-density sampling:* samples drawn from low-density regions of $p(x \mid Y=1)$ to expand the positive manifold.
 - *Noise-regularized perturbations:* small Jacobian-normal perturbations of $Y=1$ points to increase support without leaking raw data.

All feature generation is local; no raw $\mathcal{D}_k^o$ is shared.

Integration into the inner learner. Let $S_{k,t}$ denote the synthetic labeled minibatch of size m (built by joint (S, Y) stratification). We augment it with $\mathcal{U}_{k,t}^{(Y)}$ and train

$$\theta_{k,t} := \arg \min_{\theta} \left\{ \underbrace{\frac{1}{m} \sum_{(x^s, s^s, y^s) \in S_{k,t}} \ell(y^s, f_{\theta}(x^s, s^s))}_{\text{synthetic labeled}} + \lambda_U \underbrace{\frac{1}{\Delta} \sum_{(x_u^s, s_u^s, 1) \in \mathcal{U}_{k,t}^{(Y)}} \ell(1, f_{\theta}(x_u^s, s_u^s))}_{\text{Y-Universum positives (balanced } S)} + \lambda_{\theta} \frac{1}{2} \|\theta\|_2^2 \right\}. \quad (38)$$

Here $\lambda_U > 0$ is a weighting on the pseudo-positive block to avoid overfitting; λ_{θ} is ridge regularization.

Outer objective and EO constraint (unchanged form). The outer objective continues to evaluate on real data and enforce EO via the surrogate on the $Y=1$ slice:

$$\begin{aligned} \min_{\{x_j^s\}} \quad & \ell_{N_k^o}(\theta_{k,t}; \mathcal{D}_k^o) + \lambda_{\theta} \frac{1}{2} \|\theta_{k,t}\|_2^2 \\ \text{s.t.} \quad & |g_{\text{EO}}(\theta_{k,t}; \mathcal{D}_k^o)| \leq \varepsilon_{\text{EO}}. \end{aligned} \quad (39)$$

Note that $\theta_{k,t}$ depends on both $S_{k,t}$ and $\mathcal{U}_{k,t}^{(Y)}$ via (38).

Remark 1 (EO neutrality under balanced S for Y -Universum positives). *Because the added $Y=1$ pseudo-positives are split evenly across $S \in \{0,1\}$, their contribution to the EO surrogate cancels out in expectation:*

$$\mathbb{E}[g_{\text{EO}}(\theta_{k,t}; \mathcal{D}_k^o)] \approx \mathbb{E}[g_{\text{EO}}(\theta_{k,t}^o; \mathcal{D}_k^o)],$$

where $\theta_{k,t}^o$ is obtained from (38) with $\lambda_U = 0$. Thus the Y -Universum block corrects class imbalance while leaving the S -fairness constraint to be governed by the EO surrogate and AL updates.

Algorithmic insertion (client-side loop). Replace the inner step in Algorithm 2 by (38) and construct $\mathcal{U}_{k,t}^{(Y)}$ as above. The rest of the AL updates and outer synthetic-feature optimization are unchanged.

Hyperparameters. We keep the hyperparameter family $\{\lambda_\theta, \lambda, \rho, \lambda_U\}$, with $\lambda_U \in [0.01, 0.2]$ as a practical range. Set $\Delta = M - m$ per client and round, optionally capped for stability.

6.9 GAN-based Y -Universum Synthesis (Future Work)

Motivation. Beyond hand-crafted perturbations and interpolations, we can learn a generator to synthesize *pseudo-positive* samples that expand the support of $Y=1$ while preserving privacy. The goal is to produce $\Delta = M - m$ Y -Universum positives, *balanced across S* , that (i) are non-identifiable, (ii) improve minority recall, and (iii) do not bias EO fairness (by S -balancing).

Setup. Per client k , train a conditional generator $G_\phi(z, s)$ (optionally also conditioned on $y=1$) with discriminator D_ψ on local data:

$$\min_{\phi} \max_{\psi} \mathcal{L}_{\text{adv}}(G_\phi, D_\psi; \mathcal{D}_k^o) + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}}(G_\phi; \mathcal{D}_k^o) + \lambda_{\text{stab}} \mathcal{R}_{\text{stab}}(G_\phi),$$

where $\mathcal{L}_{\text{adv}}$ enforces realism, $\mathcal{L}_{\text{rec}}$ encourages coverage of $Y=1$ structure (e.g., feature-space or classifier-consistency losses), and $\mathcal{R}_{\text{stab}}$ regularizes training (e.g., gradient penalties).

Balanced synthesis. After training, draw $\Delta/2$ samples with $s=0$ and $\Delta/2$ with $s=1$:

$$\mathcal{U}_{k,t}^{(Y, \text{GAN})} = \{(x_u^s, s_u^s, 1) : x_u^s = G_\phi(z_u, s_u^s), z_u \sim p(z)\},$$

and append them to the synthetic minibatch as in (38). These are used only as $Y=1$ pseudo-positives; EO fairness with respect to S remains governed by the EO surrogate and augmented Lagrangian.

Privacy notes. GAN training and sampling occur *locally*. Only synthesized points are shared. Optionally, privacy can be strengthened via (i) noise on discriminator/gradients, (ii) membership tests to screen near-duplicates, or (iii) release thresholds based on diversity metrics.

Evaluation protocol. We will compare: (a) no Y -Universum, (b) hand-crafted Y -Universum (Sec. 6.8), and (c) GAN-based Y -Universum (this section), using:

- Minority recall/precision, macro-F1, AUROC/PR.
- EO gap (TPR difference across S on $Y=1$).
- Leakage proxies (nearest-neighbor distance to training set; diversity/coverage metrics).

Integration (toggle). In the client loop, replace $\mathcal{U}_{k,t}^{(Y)}$ with $\mathcal{U}_{k,t}^{(Y,\text{GAN})}$ when the generator is available; the inner and outer problems remain unchanged.